Detecting IP-KVM Devices in the Enterprise

From Default Indicators to Obfuscated Adversaries

A Technical White Paper for Insider Threat and Security Operations Teams

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Table of Contents

Executive Summary

IP-based Keyboard-Video-Mouse (IP-KVM) devices (small hardware appliances that provide full remote control of a computer over a network) have emerged as a significant and growing detection challenge for enterprise security teams. Originally the domain of server rooms and IT homelabs, a new generation of inexpensive, open-source IP-KVMs (PiKVM, TinyPilot, JetKVM, NanoKVM, BliKVM, and GL.iNet Comet) has attracted the attention of two distinct threat populations: **employees violating remote work policies** (the “overemployed” community), and **DPRK state-sponsored actors** operating fraudulent laptop farms on U.S. soil.

These devices operate entirely out-of-band: they connect via USB and HDMI to the target host, require no software installation, and bypass virtually all endpoint security controls by design. They present as standard keyboards, mice, and monitors to the operating system. Detecting them is possible but requires layered approaches spanning USB device telemetry, EDID display fingerprinting, network reconnaissance, and behavioral analytics.

This paper provides a complete technical primer on IP-KVM devices, catalogs their default identifiers, explains how adversaries obfuscate those identifiers, and proposes detection strategies for both unconfigured (“dumb”) and deliberately disguised (“smart”) deployments.

1. What Are IP-KVM Devices?

1.1 Fundamentals

A traditional KVM switch is a hardware device that lets you share a single keyboard, video display, and mouse across multiple computers by pressing a physical button. An IP-KVM extends this concept over a network. Instead of pressing a button at your desk, you access the target computer’s keyboard, video, and mouse through a web browser or client application from anywhere with network access.

The critical distinction: **IP-KVMs operate at the hardware layer.** From the target computer’s perspective, an IP-KVM is physically indistinguishable from a real keyboard, mouse, and monitor plugged into its ports. The target machine’s operating system sees standard USB Human Interface Devices (HID) and an HDMI display. No software, agent, or driver is installed. No process runs on the target. No network connection originates from the target. This makes IP-KVMs fundamentally different from software-based remote access tools like RDP, VNC, or TeamViewer, and fundamentally harder to detect.

1.2 How They Work

All devices in the current generation share a common architecture built on the Linux USB Gadget API (ConfigFS). A small Linux-based single-board computer sits between the operator and the target machine:

- **USB Connection (to target):** The IP-KVM's USB port connects to the target computer. Using the Linux USB Gadget subsystem, it creates a composite USB device presenting as a USB hub with one or more HID keyboards, HID mice (absolute and/or relative positioning), and optionally a USB mass storage device, Ethernet adapter, or serial port.
- **HDMI Connection (from target):** The IP-KVM captures video output via HDMI. It appears to the target as a connected display, sending EDID (Extended Display Identification Data) to negotiate resolution and capabilities. Captured video is encoded (H.264/H.265) and streamed to the operator.
- **Network Connection (to operator):** The IP-KVM connects to the network via Ethernet or Wi-Fi. The operator accesses a web interface or connects via Tailscale, WireGuard, or cloud management.
- **Optional Extras:** Virtual media mounting (ISOs as USB drives), serial console, Wake-on-LAN, mouse jiggling (to prevent screen lock), and ATX power control.

1.3 The Current Device Landscape

The market has exploded since 2020. Six primary devices dominate:

Device	Platform	Price	Default Auth	Key Feature
PiKVM	Raspberry Pi	$230–$400+	Yes	The original; mature, well-audited
TinyPilot	Raspberry Pi	$399	No (free); Yes (Pro)	Commercial; polished UX
JetKVM	Rockchip ARM	$69	Setup-configurable	Cloud-managed; UI-based profile spoofing
NanoKVM	RISC-V	$25–$100	Yes	Ultra-cheap; serious security issues
BliKVM	AllWinner / RPi	$110–$300	Varies by OS	Can run PiKVM OS or native stack
GL.iNet Comet	PiKVM fork	~$90	Yes	Travel-router company's KVM entry

According to runZero research (June 2025), hundreds of PiKVM devices are exposed on the public internet, followed by NanoKVM and TinyPilot in the dozens-to-low-hundreds range, with JetKVM, BliKVM, and GL.iNet Comet in smaller numbers. Internally, PiKVM and JetKVM are the most frequently encountered in corporate environments, though legacy enterprise KVMs (Avocent, Raritan) remain roughly 10x more prevalent in large organizations. The market is growing quickly. JetKVM's December 2024 Kickstarter raised over $4.3 million from 31,000+ backers.

2. Why IP-KVMs Are a Risk

2.1 The Overemployed Use Case

The r/overemployed community on Reddit (individuals holding multiple simultaneous remote jobs) has adopted IP-KVMs as essential infrastructure. The use case: leave company-issued laptops at home connected to IP-KVMs, then control them remotely from anywhere, making it appear the employee is working locally. Built-in mouse jigglers prevent screen lock, simulating continuous presence.

A May 2025 Reddit post titled “Corporate detected my IP KVM today” (551 upvotes, 218 comments) catalyzed significant community discussion. The poster was using a TinyPilot in its default configuration and was caught within a year. Comments from cybersecurity professionals confirmed that CrowdStrike Falcon and other EDR platforms now actively scan for these devices. One user stated: *“There’s a couple other tricks we use to identify [IP-KVMs] that isn’t covered by the documentation, but I’d rather not post them on a public forum.”*

A January 2026 thread, “KVM getting banned by companies,” indicated a broader enterprise trend of implementing detection and blocking policies. The overemployed community represents an ongoing detection challenge because users are motivated to evade, share evasion techniques openly, and are typically technically sophisticated.

2.2 The DPRK Laptop Farm Threat

Far more consequential is use by North Korean state-sponsored threat actors. CrowdStrike tracks this as **Famous Chollima** (Microsoft: **Jasper Sleet**). The operation is well-documented:

- DPRK IT workers obtain employment at U.S. companies using stolen or fabricated identities, often with AI-assisted interview coaching.
- Company-issued laptops are shipped to U.S.-based accomplices operating laptop farms, rooms containing dozens of devices, each connected to a PiKVM or TinyPilot.
- DPRK operators connect from overseas via Tailscale mesh VPN or Astrill VPN, controlling machines through the IP-KVMs.
- Revenue is funneled through shell companies and fake bank accounts back to North Korea, supporting WMD and ballistic missile programs.

The scale is staggering. CrowdStrike’s 2024 report documented **304 Famous Chollima incidents**, with nearly 40% representing insider threat operations. In June–July 2025, the DOJ’s Operation “DPRK RevGen: Domestic Enabler Initiative” searched **29 suspected laptop farms across 16 states**, seizing financial accounts and fake websites. More than 100 U.S. companies had been infiltrated, including Fortune 500 organizations. Sensitive data including U.S. military technology regulated under ITAR was accessed and exfiltrated.

Nisos published a March 2026 account of discovering a laptop farm after a DPRK actor targeted their own company. Through a compromised laptop’s camera, they captured images of a closet containing approximately 40 devices, 20 actively part of the farm — all controlled via PiKVM with Tailscale mesh VPN.

2.3 The Broader Security Risk

Even absent malicious intent, IP-KVMs introduce significant risk:

- **Bypass of endpoint controls:** EDR, DLP, and application allowlisting are irrelevant because the device operates below the OS layer.
- **Pre-boot access:** An attacker who compromises the KVM can access BIOS/UEFI, boot from external media, and bypass full-disk encryption.
- **Screen lock bypass:** Built-in mouse jigglers defeat idle timeout policies.
- **Virtual media:** Many KVMs can mount ISO images as USB drives, enabling data exfiltration or malware delivery without touching the network.
- **Network exposure:** A misconfigured KVM exposed to the internet provides unauthenticated access. A 2024 HackerOne report documented a DoD workstation fully exposed via a TinyPilot with no authentication.
- **Lateral movement:** If a KVM is compromised on an employee's home network, it provides a direct path to the enterprise device.

3. Detecting the "Dumb" Ones: Default Indicators

The good news: most IP-KVMs ship with predictable, static, and easily searchable default identifiers. Users who don't modify these defaults (the majority) are trivially detectable with the right telemetry.

3.1 USB Device Identifiers

All current-generation IP-KVMs register in the operating system's USB device tree with identifiable default attributes. The following indicators are sourced from hands-on device testing by Grumpy Goose Labs (wav3, November 2025) and validated against PiKVM, runZero, and SANS ISC documentation.

PiKVM (also Geekworm KVM-A3/A4/A8/X650, Pi-Cast)

Attribute	Value
Vendor ID (VID)	1D6B (The Linux Foundation)
Product ID (PID)	0104 (Multifunction Composite Gadget)
Serial Number	CAFEBABE
Manufacturer String	PiKVM
Product String	Composite KVM Device
Configuration Descriptor	Config 1: PiKVM
MaxPowerDraw	125
NumInterfaces	3
DeviceInstanceId	USB\VID_1D6B&PID_0104\CAFEBABE

Observed OUI	28:CD:C1, 2C:CF:67, D8:3A:DD, DC:A6:32, E4:5F:01, and others (Raspberry Pi Foundation)

TinyPilot Voyager 2A

Attribute	Value
Vendor ID (VID)	1D6B
Product ID (PID)	0104
Serial Number	6b65796d696d6570690
Manufacturer String	tinypilot
Product String	Multifunction USB Device
Configuration Descriptor	Config 1: ECM network
MaxPowerDraw	125
NumInterfaces	2
DeviceInstanceId	USB\VID_1D6B&PID_0104\6b65796d696d6570690
Observed OUI	D8:3A:DD (Raspberry Pi Foundation)
Audio Device	“TinyPilot” (persists after EDID change)

GL.iNet Comet (GLKVM)

Attribute	Value
Vendor ID (VID)	1D6B
Product ID (PID)	0104
Serial Number	CAFEBABE
Manufacturer String	GLKVM (v1.3) / Glinet (v1.5+)
Product String	Composite KVM Device
Configuration Descriptor	Config 1: GLKVM device (v1.3) / Glinet device (v1.5+)
MaxPowerDraw	125
NumInterfaces	4 (v1.3), 5 (v1.5), 7 (with obfuscation)
DeviceInstanceId	USB\VID_1D6B&PID_0104\CAFEBABE
Observed OUI	94:83:C4 (GL.iNet)
Audio Device	“GLKVM” + Microphone Source/Sink
CRITICAL NOTE	All 4 built-in obfuscation profiles retain serial CAFEBABE

JetKVM

Attribute	Value

Vendor ID (VID)	1D6B
Product ID (PID)	0104
Serial Number	(empty by default)
Manufacturer String	JetKVM
Product String	JetKVM USB Emulation Device
Configuration Descriptor	Config 1: HID
MaxPowerDraw	125
NumInterfaces	4
DeviceInstanceId	USB\VID_1D6B&PID_0104\CAFEBABE
Observed OUI	80:34:28

NanoKVM (Sipeed)

Attribute	Value
Vendor ID (VID)	3346 (Sipeed)
Product ID (PID)	1009
Serial Number	0123456789ABCDEF
Manufacturer String	sipeed
Product String	NanoKVM
Configuration Descriptor	NanoKVM
MaxPowerDraw	60
NumInterfaces	6
DeviceInstanceId	USB\VID_3346&PID_1009\0123456789ABCDEF
Observed OUI	48:DA:35:60:*

BliKVM

Attribute	Value
Vendor ID (VID)	1D6B
Product ID (PID)	0106
Serial Number	6b65796d696d6570690
Manufacturer String	BliKVM
Product String	Multifunction
Configuration Descriptor	Config 1: ECM network
MaxPowerDraw	125
NumInterfaces	4

DeviceInstanceId	USB\VID_1D6B&PID_0106\6b65796d696d6570690
Observed OUI	12:00:0f:*

Openterface Mini-KVM

Attribute	Value
Vendor ID (VID)	1A86 (WCH / Nanjing Qinheng)
Product ID (PID)	E329
Serial Number	2019B152ED98
Manufacturer String	WWW.WCH.CN
Product String	WCH UART TO KB-MS_V1.8
MaxPowerDraw	50
NumInterfaces	4
DeviceInstanceId	USB\VID_1A86&PID_E329\2019B152ED98

Most devices share VID `1D6B` (Linux Foundation) and PID `0104`, but NanoKVM uses Sipeed's own VID `3346`, BliKVM uses PID `0106`, and Openterface uses a WCH chip with VID `1A86`. The combination of VID/PID with distinctive manufacturer strings, serial numbers, or configuration descriptors is highly indicative. Additional PiKVM-derivative brands (Geekworm, Pi-Cast) run unmodified PiKVM OS and inherit identical USB indicators, though their hardware OUI and mDNS hostnames may differ.

Detection Queries

CrowdStrike Falcon (default state):

```
#event_simpleName=/^DcUsbDevice(Connected|Disconnected)$/ /kvm/i
```

CrowdStrike Falcon (PiKVM serial):

```
#event_simpleName=/^DcUsbDevice(Connected|Disconnected)$/ CAFEBABE
```

Windows PowerShell (PiKVM):

```
Get-CimInstance -ClassName Win32_PnPEntity |
  Where-Object { $_.PNPDeviceID -like '*CAFEBABE*' } |
  Select-Object Description, Name, DeviceID, Manufacturer
```

Splunk (generic EDR USB logs):

```
index=edr sourcetype=usb_events
  (manufacturer="*pikvm*" OR manufacturer="*tinypilot*"
   OR manufacturer="*jetkvm*" OR manufacturer="*nanokvm*"
   OR manufacturer="*sipeed*" OR manufacturer="*blikvm*"
   OR manufacturer="*glkvm*" OR manufacturer="*glinet*"
```

```
   OR manufacturer="*WCH*"
   OR serial="CAFEBABE" OR serial="6b65796d696d6570690"
   OR serial="0123456789ABCDEF"
   OR vid="3346" OR vid="1A86")
 | stats count by host, manufacturer, serial, vid, pid
```

CrowdStrike Falcon (comprehensive, all known brands):

```
#event_simpleName = DcUsb*
/tinypilot/i OR /pikvm/i OR /aurga/i OR /blikvm/i
  OR /geekworm/i OR /jetkvm/i OR /nanokvm/i OR /picast/i
groupBy([ComputerName, ConfigurationDescriptorName,
         DeviceManufacturer, DeviceSerialNumber])
```

This query uses the wildcard DcUsb* event type to catch all USB-related events (connections, disconnections, configuration changes) and groups by endpoint name, configuration descriptor, manufacturer string, and serial number. The regex pattern covers PiKVM, TinyPilot, Aurga, BliKVM, Geekworm, JetKVM, NanoKVM, and PiCast variants.

A note on vendor research: Okta Threat Intelligence (Alex Tilley, April 2025) confirmed that their team independently tested multiple IP-KVM devices and developed detection approaches. The resulting observables are available to Okta customers through the authenticated portal at security.okta.com. Their public advisory noted that, at the time of publication, no EDR vendor provided signatures specifically designed to detect IP-KVM devices, making multi-layered detection essential.

3.2 EDID Display Identifiers

When an IP-KVM captures HDMI output, it presents itself as a connected display using EDID data. Default values vary significantly by device and are highly distinctive:

Device	EDID Mfr	Instance	FriendlyName	Resolution	Refresh
PiKVM	LNX	DISPLAY\LNX777X	PiKVM V3	1920x1080	30Hz
TinyPilot	TSB	DISPLAY\TSB9876	TinyPilot	1920x1080	30Hz
GL.iNet Comet	GLI	DISPLAY\GLIC21C	GLIKVM	2560x1440	59Hz
JetKVM	TSB	DISPLAY\TSB8801	T749-fHD720	1920x1080	60Hz
NanoKVM	VCS	DISPLAY\VCS1145	Connector	1920x1080	60Hz
BliKVM	HJW	DISPLAY\HJW0001	HDMI TO USB	1920x1080	60Hz
Openterface	HJW	DISPLAY\HJW0001	HD TO USB	1920x1080	60Hz

Several patterns are notable. PiKVM uses the `LNX` manufacturer code (Linux Foundation), a synthetic EDID that no legitimate display uses. TinyPilot and JetKVM both default to `TSB` (Toshiba) codes referencing discontinued monitors from 2011-2020. GL.iNet Comet uses its own `GLI` code. BliKVM and Openterface share the `HJW` code, likely from a common HDMI capture chip. NanoKVM uses `VCS` (Sipeed's registered code). Any endpoint reporting a display with these manufacturer codes, especially combined with suspiciously low refresh rates (30Hz) or discontinued monitor friendly names, warrants investigation.

On Windows, check `Microsoft-Windows-DeviceSetupManager/Admin` Event ID 112 for display connection events. The `WmiMonitorID` WMI class provides ManufacturerName, ProductCodeID, SerialNumberID, and UserFriendlyName for all connected displays.

3.3 Network-Based Detection

IP-KVMs expose fingerprintable services on the network:

- **Web interface titles:** HTML <title> tags containing PiKVM, TinyPilot, JetKVM, etc.
- **Favicon hashes (Shodan):** PiKVM: -1040945478; TinyPilot: -996415781; JetKVM: -1261329937; NanoKVM: 1323732765
- **TLS certificates:** Static or predictable X.509 subject/issuer fields in self-signed certs.
- **mDNS hostnames:** pikvm.local, tinypilot.local, jetkvm-XXXX.local
- **MAC address OUI:** Raspberry Pi Foundation, Sipeed/Sispeed, GL.iNet prefixes identify underlying hardware.
- **Shodan correlation:** Feed remote worker public IPs from VPN logs into Shodan/Censys to find exposed KVM interfaces.

4. Detecting the “Smart” Ones: Obfuscated Deployments

This is where things get hard. A motivated adversary can modify every default identifier discussed in Section 3. The question becomes: what signals survive obfuscation?

4.1 How Identifiers Are Changed

PiKVM: The `/etc/kvmd/override.yaml` file allows changing VID, PID, serial number, manufacturer string, and product string. EDID can be customized via a separate configuration. Changes survive reboots but may be overwritten by OS updates.

TinyPilot: The `/opt/tinypilot-privileged/init-usb-gadget` file controls USB descriptors. TinyPilot’s own FAQ explains how to modify identifiers while noting changes are “not supported” and will be overwritten during updates.

JetKVM: The most evasion-friendly device. JetKVM provides a **UI-based dropdown menu** for changing device profiles directly in the web interface. The frontend includes preset configurations that mimic common USB devices: select a Logitech profile with a few clicks. Custom VID/PID/serial/manufacturer strings can be set through the configuration.

NanoKVM: ConfigFS parameters can be modified, though the process is less documented and the device’s poor software quality means changes may not persist reliably.

4.2 The Firmware Update Problem

Firmware and software updates frequently reset customized identifiers to factory defaults. This is a critical and underappreciated vulnerability in adversary evasion strategies.

In the Sazkwatch case study (August 2025), a corporate analyst found seven users running PiKVM or TinyPilot devices. One user had disguised their PiKVM as a Corsair keyboard, but a firmware update reset the device to factory defaults, re-exposing the CAFEBABE serial number. Three separate users were caught in a “cat-and-mouse” game of modifying identifiers, having them reset by updates, and modifying again.

Detection opportunity: Retrospective analysis of USB device logs may reveal transient exposures of default identifiers between update cycles, even for otherwise well-disguised devices. A device that appeared as “PiKVM” for even a few minutes before reverting to “Logitech Keyboard” is a strong investigative lead. This requires retaining USB telemetry over time.

4.3 Residual Indicators That Survive Obfuscation

Even when an adversary customizes all obvious identifiers, several signals persist:

4.3.1 USB Composite Device Structure

IP-KVMs present as USB composite devices. Even when individual device names are spoofed, the structure of the composite device is distinctive:

- **Child device count:** A PiKVM presents 3–5 child devices (keyboard, absolute mouse, relative mouse, optional mass storage/Ethernet). A real keyboard or mouse is a single device. A hub with exactly one keyboard + one absolute mouse + one relative mouse on a single USB port is anomalous.
- **Absolute mouse positioning:** Real USB mice report relative movement (deltas). IP-KVMs typically use absolute positioning (like a graphics tablet) to avoid cursor sync issues. The presence of both absolute and relative mice on the same hub is a strong indicator.
- **Power draw characteristics:** The power budget of a KVM’s emulated device tree differs from real peripherals. This data is available in USB enumeration events.
- **DeviceDescriptorSetHash:** CrowdStrike Falcon generates a hash of USB device descriptor attributes. Even when individual fields change, identical configurations produce identical hashes, useful for identifying a batch of identically-configured devices in a laptop farm.

4.3.2 DcUsbConfigurationDescriptor Events

The CrowdStrike event `DcUsbConfigurationDescriptor` captures the device’s configuration descriptor string, a field that often retains identifying information even after partial obfuscation. An adversary who changes the manufacturer string but forgets the configuration descriptor (set in a different location) is still detectable.

4.3.3 EDID Anomalies

Even if EDID strings are changed, anomalies may persist:

- **Resolution mismatch:** Limited or odd resolution support unlike any real modern display.
- **Vintage identification:** EDID codes assigned to long-discontinued monitors (e.g., Toshiba LCDs from 2007) on a brand-new laptop.
- **Missing capabilities:** Real modern displays advertise HDR, wide color gamut, etc. KVM EDIDs are minimal.
- **Audio device persistence:** On Windows, TinyPilot registers an audio device named "TinyPilot" that persists even after EDID changes.

4.3.4 MAC Address and Network Fingerprinting

MAC addresses are burned into hardware and significantly harder to spoof than USB descriptors. Raspberry Pi Foundation OUI prefixes identify the underlying hardware. CrowdStrike Falcon's Unmanaged Assets feature can detect Raspberry Pi devices on the same LAN segment as corporate endpoints.

4.3.5 Behavioral Indicators

When all hardware identifiers are successfully spoofed, behavioral analysis becomes the last line of defense:

– **Input pattern analysis:** Mouse movements through a KVM have different timing characteristics due to network latency and video encoding. EDR platforms have indicated capability to detect non-human patterns.

– **Session duration:** A machine that never sleeps, never locks, and shows continuous low-level activity 24/7 is suspicious.

– **Keyboard cadence:** KVM-mediated typing introduces latency and keystroke batching detectable by statistical analysis of inter-keystroke timing.

– **Resolution changes:** KVM operators resizing their browser trigger display resolution renegotiation, unusual for physically-connected monitors.

4.3.6 Cross-Correlation: Identifier Change Events

A particularly powerful technique: **watching for the same physical USB port showing different device identifiers across time.** If a port shows "PiKVM" on Monday, then "Logitech" on Tuesday with the same composite structure, that is a strong indicator of configuration changes. As one CrowdStrike administrator confirmed: "It shows up as changed from whatever it was originally. Logs are preserved so it's easy to pull."

4.3.7 VID/PID and Manufacturer Brand Mismatch

When adversaries spoof USB device identifiers, they frequently introduce inconsistencies that a real device would never have. A device presenting itself as a Logitech product, for example, should carry a Logitech VID and a manufacturer string of "Logitech." If the device claims to be a

Logitech mouse but lists “Microsoft” (or any other brand) in its manufacturer field, that mismatch is itself an indicator of concern.

More broadly, any USB device where the VID (which is assigned by the USB Implementers Forum to a specific company) does not correspond to the manufacturer string, or where the product string names a product from a different vendor than the VID indicates, warrants investigation. Real hardware is manufactured with consistent branding across all descriptor fields. A spoofed device assembled from mix-and-matched values across different brands is a telltale sign of manual ConfigFS configuration.

This check is straightforward to implement: maintain a lookup table mapping VIDs to their registered manufacturer names, then flag any USB device events where the manufacturer string contradicts the VID. False positives are rare because legitimate hardware vendors control both values at the factory.

4.4 Advanced Adversary Techniques

The most sophisticated adversaries employ additional countermeasures:

– **EDID pass-through dongles:** Small HDMI adapters that present their own EDID instead of the KVM’s. Defeats EDID detection but adds another identifiable device.

– **USB hub interposition:** Placing a docking station between KVM and target masks the KVM’s USB device tree. Effectiveness depends on hub pass-through behavior.

– **VGA instead of HDMI:** Eliminates EDID detection entirely but limits video quality.

– **Network isolation:** Placing the KVM on a separate VLAN or cellular connection prevents network-based discovery. Doesn’t affect host-based USB/EDID detection.

– **Custom HID builds:** Man-in-the-middle USB gadgets built from microcontrollers (Arduino, RP2040) that perfectly mimic legitimate hardware. Detectable only through behavioral analysis or physical inspection.

5. Building a Detection Program

5.1 Maturity Model

Tier	Approach	What It Catches	Effort
Tier 1	Default indicator matching	Unconfigured devices (majority)	Immediate: query existing EDR/SIEM
Tier 2	Structural analysis	Partially obfuscated devices	Short-term: composite device tree rules
Tier 3	Temporal correlation	Devices reset by firmware updates	Medium-term: historical log retention
Tier 4	Behavioral analytics	Fully disguised devices	Long-term: ML/statistical models

5.2 Framework Mapping

Both major frameworks have formalized IP-KVM detection:

MITRE ATT&CK

- **Technique:** T1219.003 — Remote Access Tools: Remote Access Hardware
- **Detection Strategy:** DET0159 — Detect Remote Access via USB Hardware
- **Analytics:** AN0446, AN0447, AN0448 (Windows, Linux, macOS)

Insider Threat Matrix™ (ITM)

The Insider Threat Matrix™, developed by Forscie and maintained by James Weston and Joshua Beaman, categorizes IP-KVM deployment as **PR036 — Hardware-Based Remote Access (IP-KVM)** under **AR3 (Preparation)**. Created March 2026 (contributor: Leonardo Segura), the entry has 10 linked preventions and 14 linked detections. The ITM classifies this as *preparatory activity,* establishing a covert remote control capability that may later enable unauthorized access, data exfiltration, or system manipulation. This is distinct from MITRE's treatment, which focuses on the execution-phase technique rather than investigative framing.

The ITM's emphasis on investigative trajectory (mapping behavior from motive through preparation to infringement) makes it particularly useful for insider threat teams building cases, while MITRE ATT&CK's detection strategy (DET0159) is more directly actionable for SOC detection engineering.

5.3 Required Data Sources

Data Source	Used For	Priority
EDR USB device events	VID/PID/serial matching, composite analysis	Critical
Windows Event ID 112	USB and display connection events	High
EDID data (WMI WmiMonitorID)	Display fingerprinting	High
Network asset inventory	Network-based device discovery	High
mDNS/DNS logs	Hostname discovery	Medium
DHCP logs	MAC to hostname correlation	Medium
VPN/proxy logs	External IP correlation for Shodan	Medium
EDR Unmanaged Assets	LAN device discovery	Medium
Input behavioral telemetry	Timing analysis	Low (high value)

5.4 Policy Recommendations

- **Asset policy:** Define whether IP-KVM devices are authorized. Add them to prohibited device lists with justification.

- **USB allowlisting:** Where possible, implement USB device allowlisting via EDR. Block unknown composite USB devices by default.
- **Remote worker inventory:** Require remote workers to report all devices connected to company equipment.
- **EDID baselining:** Baseline displays per endpoint. Alert on new or changed displays with unusual EDID data.
- **Log retention:** Retain USB device logs for at least 12 months. Transient KVM exposures during firmware updates may be the only detection window.
- **Incident response playbook:** Develop specific procedures for KVM detection events: authorized use check, DPRK indicators (multiple LAN devices, Tailscale/Astrill traffic), insider threat team engagement.

6. Looking Ahead

The detection landscape is evolving rapidly. JetKVM already provides UI-based profile spoofing with presets mimicking legitimate peripherals. The overemployed community shares evasion techniques openly, and DPRK operators demonstrate willingness to adapt as detection improves.

At the same time, EDR vendors are adding detection capabilities. CrowdStrike updated Falcon to block PiKVM USB input on protected endpoints as of August 2024. MITRE's formalization of T1219.003 provides standardized language driving broader vendor adoption.

Emerging countermeasures include USB device attestation (cryptographic verification), HDMI authentication leveraging HDCP, machine learning on input telemetry, and physical security integration. None are mature today, but the trajectory is clear.

The fundamental insight: IP-KVMs are **hardware implants** that happen to be commercially available. Effective detection requires **multiple overlapping approaches**: host-based telemetry, network reconnaissance, temporal analysis, and behavioral analytics. No single technique is sufficient, but the combination creates a detection mesh that is increasingly difficult for adversaries to fully evade.

The "dumb" ones are easy. The "smart" ones are hard. But even the smart ones leave traces — the art is knowing where to look.

7. The Expanding Landscape: Emerging and Upcoming Devices

The IP-KVM market is expanding rapidly. Beyond the six primary devices covered in this paper, a wave of new entrants is entering the market through crowdfunding platforms, direct sales, and Amazon/AliExpress listings. Each new device represents a potential detection gap if security teams are only tuned to the current generation's signatures.

7.1 LeafKVM (Crowd Supply, April 2026)

LeafKVM launched on Crowd Supply on April 21, 2026, with a $119 price point and a $10,000 funding goal (82% funded within the first week). Built on the Rockchip RV1126B SoC (quad-core Cortex-A53, 512 MB RAM), it features Wi-Fi 5, PoE support, a 2.4-inch touchscreen, and USB-C serial debug console. The backend is written entirely in Rust, and the web frontend is a **GPL-2.0 fork of the JetKVM project**. Shipping is scheduled for mid-January 2027.

Detection relevance: Because LeafKVM's frontend is forked from JetKVM, it likely shares similar USB gadget defaults (VID 1D6B, PID 0104) unless the developers customize them. As a fully open-source project (hardware schematics included), it will be trivially modifiable. Its Rust backend is a departure from the Python (PiKVM) and Go (JetKVM/NanoKVM) stacks seen elsewhere, potentially producing different system-level artifacts. Security teams should monitor for Rockchip-based MAC addresses and any new default identifiers once the firmware is published.

7.2 GL.iNet Comet Product Line Expansion

GL.iNet, already in the market with the original Comet (PiKVM fork), is aggressively expanding:

- **Comet Pro (GL-RM10):** Wi-Fi 6, dedicated mobile apps (Windows/macOS/iOS/Android), cloud management via glkvm.com, Tailscale integration. Available on Amazon and AliExpress.
- **Comet 5G (GL-RM10RC):** Adds 5G RedCap cellular connectivity with 4G LTE fallback, 3.69-inch touchscreen, and independent out-of-band management channel. This is particularly concerning from a detection standpoint; the cellular connection means the KVM can operate entirely outside the corporate network, making network-based detection impossible.
- **Comet Q (GL-RMQ1):** A USB-C KVM designed for phones and tablets, the first KVM to support controlling iPhones. Operates over Wi-Fi rather than HDMI capture. Represents a new form factor that traditional IP-KVM detection signatures won't cover.

Detection relevance: The Comet line runs GLKVM OS (Buildroot-based Linux). Because it's based on PiKVM's kvmd, many of the same USB gadget defaults (VID 1D6B, PID 0104, LNX EDID) apply. However, the Comet 5G's cellular connectivity creates a genuinely new detection challenge; the management channel never touches the corporate network, so all detection must occur at the host USB/EDID layer.

7.3 Openterface Mini-KVM and KVM-GO

Openterface has carved a niche with **KVM-over-USB** devices, a different architecture than IP-KVMs. Rather than connecting to a network, the Openterface Mini-KVM ($461K on Crowd Supply) and KVM-GO ($119, launched November 2025) connect directly between two computers via USB. Your laptop becomes the console for the target machine. While these devices don't provide remote network access, they enable the same kind of BIOS-level control and could be used in an office environment to access a locked-down workstation from a personal laptop.

Detection relevance: Because Openterface devices use USB-C direct connection rather than network connectivity, they won't appear in network scans. They still present as USB HID devices to the target and will have their own default VID/PID/serial signatures. Their small form factor (the KVM-GO is keychain-sized) makes physical detection challenging.

7.4 The Amazon/AliExpress Factor

IP-KVMs are increasingly available as commodity purchases on Amazon and AliExpress. The GL.iNet Comet and Comet Pro are both listed on Amazon with Prime shipping. NanoKVM devices are available on AliExpress for as little as $25. JetKVM continues to take orders via its Kickstarter page. Numerous unbranded Chinese IP-KVM clones are also appearing on these platforms, often based on the same Rockchip or Allwinner SoCs with modified PiKVM or NanoKVM firmware.

Detection relevance: Unbranded clones are a growing concern. They may ship with unpredictable default identifiers, unknown firmware, and potentially pre-installed backdoors. Unlike the well-documented devices covered earlier, these clones may not have published default VID/PID/serial values, making signature-based detection unreliable. Structural and behavioral detection techniques (Tier 2–4) become essential for catching these devices.

7.5 Implications for Detection Teams

The proliferation of IP-KVM devices (now spanning at least a dozen distinct products across multiple hardware platforms, with prices ranging from $25 to $400+) means that maintaining a static list of known-bad USB identifiers is necessary but not sufficient. Detection programs must invest in the structural and behavioral techniques described in Section 4 to remain effective as the device landscape continues to fragment.

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Appendix A: Quick-Reference Detection Cheat Sheet

Indicator	Type	Catches Default?	Catches Obfuscated?	Data Source
Serial = CAFEBABE	USB	Yes (PiKVM/GLKVM/JetKVM)	No	EDR USB events
Serial = 6b65796d696d6570690	USB	Yes (TinyPilot/BliKVM)	No	EDR USB events
Serial = 0123456789ABCDEF	USB	Yes (NanoKVM)	No	EDR USB events
VID=3346 PID=1009	USB	Yes (NanoKVM only)	Partial	EDR USB events
VID=1A86 PID=E329	USB	Yes (Openterface)	Partial	EDR USB events
Manufacturer contains 'kvm'	USB	Yes (most)	No	EDR USB events
VID=1D6B + PID=0104 + composite	USB	Partial	Partial	EDR USB events
EDID: LNX/TSB/GLI/VCS/HJW	Display	Yes (per device)	No	WMI, Event 112
GLKVM obfuscation + CAFEBABE	USB	Yes (GL.iNet)	Yes!	EDR USB events
Raspberry Pi / GL.iNet OUI on LAN	Network	Partial	Yes	EDR Unmanaged Assets
Absolute + relative mouse on hub	Structural	No	Yes	EDR USB events
USB ID change on same port	Temporal	No	Yes	Historical EDR logs
VID/PID vs manufacturer mismatch	USB	No	Yes	EDR USB events
Non-human mouse patterns	Behavioral	No	Yes	Advanced EDR
Machine never sleeps/locks	Behavioral	No	Yes	Endpoint telemetry
Transient default ID (updates)	Temporal	No	Yes	Historical EDR logs
Config desc: ECM|PiKVM|HID|GLKVM	USB	Yes	Partial	CrowdStrike
Audio device name mismatch	Audio	Yes	Partial	Endpoint audio enum

Appendix B: Default Identifiers: Complete Reference

Source: Grumpy Goose Labs, “Be KVM, Do Fraud” (wav3, November 2025), validated against PiKVM/runZero/SANS ISC documentation. Firmware versions as of November 2025.

PiKVM (and Geekworm/Pi-Cast derivatives)

```
VID:            1D6B (Linux Foundation)
PID:            0104 (Multifunction Composite Gadget)
Serial:         CAFEBABE
Manufacturer:   PiKVM
Product:        Composite KVM Device
Config Desc:    Config 1: PiKVM
MaxPowerDraw:   125
NumInterfaces:  3
InstanceId:     USB\VID_1D6B&PID_0104\CAFEBABE
EDID Mfr:       LNX  |  Instance: DISPLAY\LNX777X
EDID Name:      PiKVM V3  |  1920x1080 @ 30Hz
Observed OUI:   28:CD:C1, 2C:CF:67, D8:3A:DD, DC:A6:32, E4:5F:01
Config File:    /etc/kvmd/override.yaml
```

TinyPilot Voyager 2A

```
VID:            1D6B  |  PID: 0104
Serial:         6b65796d696d6570690
Manufacturer:   tinypilot
Product:        Multifunction USB Device
Config Desc:    Config 1: ECM network
MaxPowerDraw:   125  |  NumInterfaces: 2
InstanceId:     USB\VID_1D6B&PID_0104\6b65796d696d6570690
EDID Mfr:       TSB  |  Instance: DISPLAY\TSB9876
EDID Name:      TinyPilot  |  1920x1080 @ 30Hz  |  Yr: 2020
Audio Device:   "TinyPilot" (persists after EDID change)
Observed OUI:   D8:3A:DD
```

GL.iNet Comet (GLKVM)

```
--- v1.3.0 defaults ---
VID: 1D6B  |  PID: 0104  |  Serial: CAFEBABE
Manufacturer: GLKVM  |  Config Desc: Config 1: GLKVM device
NumInterfaces: 4

--- v1.5.0 defaults ---
Manufacturer: Glinet  |  Config Desc: Glinet device
NumInterfaces: 5 (or 7 with obfuscation profiles)

EDID Mfr:       GLI  |  Instance: DISPLAY\GLIC21C
EDID Name:      GLIKVM  |  2560x1440 @ 59Hz  |  Yr: 2021
Audio:          "GLKVM" + Microphone Source/Sink
Observed OUI:   94:83:C4

BUILT-IN OBFUSCATION (all retain serial CAFEBABE!):
  #1 Logitech  VID_046D&PID_C526  EDID: ASUS ROG PG248Q
  #2 Corsair   VID_6940&PID_6973  EDID: ViewSonic VX2478-2
  #3 Dell      VID_413C&PID_2011  EDID: Lontium semi 4K
  #4 Microsoft VID_045E&PID_005F  (keyboard only)
```

JetKVM (APP 0.4.8 / System 0.2.5)

```
VID: 1D6B  |  PID: 0104  |  Serial: (empty by default)
Manufacturer: JetKVM  |  Product: JetKVM USB Emulation Device
Config Desc:    Config 1: HID
MaxPowerDraw:   125  |  NumInterfaces: 4
EDID Mfr:       TSB  |  Instance: DISPLAY\TSB8801
```

```
EDID Name:      T749-fHD720  |  1920x1080 @ 60Hz  |  Yr: 2011
Audio:          "T749-fHD720"
Observed OUI:   80:34:28

EDID obfuscation presets:
  #1 Acer B246WL      #2 ASUS PA248QV
  #3 DELL D2721H      #4 DELL iDRAC (1280x1024, Yr: 2007)
```

NanoKVM (Sipeed) - v2.2.0 / v2.2.9

```
VID:            3346 (Sipeed)  *** UNIQUE VID ***
PID:            1009
Serial:         0123456789ABCDEF
Manufacturer:   sipeed  |  Product: NanoKVM
Config Desc:    NanoKVM
MaxPowerDraw:   60  |  NumInterfaces: 6
InstanceId:     USB\VID_3346&PID_1009\0123456789ABCDEF
EDID Mfr:       VCS  |  Instance: DISPLAY\VCS1145
EDID Name:      Connector  |  1920x1080 @ 60Hz  |  Yr: 2021
Observed OUI:   48:DA:35:60:*
Security Grade: F (runZero, June 2025)
```

BliKVM (v2.2.1-alpha, native stack)

```
VID: 1D6B  |  PID: 0106  *** DIFFERENT PID ***
Serial:         6b65796d696d6570690  (same as TinyPilot)
Manufacturer:   BliKVM  |  Product: Multifunction
Config Desc:    Config 1: ECM network
MaxPowerDraw:   125  |  NumInterfaces: 4
EDID Mfr:       HJW  |  Instance: DISPLAY\HJW0001
EDID Name:      HDMI TO USB  |  1920x1080 @ 60Hz  |  Yr: 2019
Observed OUI:   12:00:0f:*
NOTE: When running PiKVM OS, uses PiKVM defaults instead
```

Openterface Mini-KVM (KVM-over-USB, not IP)

```
VID: 1A86 (WCH)  |  PID: E329  *** UNIQUE VID ***
Serial:         2019B152ED98
Manufacturer:   WWW.WCH.CN
Product:        WCH UART TO KB-MS_V1.8
MaxPowerDraw:   50  |  NumInterfaces: 4
EDID Mfr:       HJW  |  Instance: DISPLAY\HJW0001
EDID Name:      HD TO USB  |  1920x1080 @ 60Hz  |  Yr: 2019
Audio:          "HD TO USB"
```
